

HIKET — Bayesian Calibration

Methods and pipeline documentation

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1 Overview

This document describes the Bayesian calibration pipeline for the HIKET multi-model SOC intercomparison study. The calibration is implemented in a model-agnostic engine (`calibration_engine.R`) and five model-specific orchestration scripts (`run_{Model}_calibration.R`).

Models in the intercomparison: SP1, TP2, Yasso07, Yasso15, Yasso20.

Model complexity spectrum. The five models form a deliberate complexity ladder: SP1 (1 pool, 6 free parameters), TP2 (2 pools, 8 free parameters), and Yasso07/15/20 (5 pools, 15–26 free parameters depending on version). This design allows the question “how much structural complexity does Finnish SOC data actually require?” to be answered quantitatively, with all models calibrated under the same statistical framework against the same data.

Why not RothC? RothC was originally planned as the fifth comparison model but was replaced by SP1 and TP2. RothC introduces conceptual differences beyond pool count — clay-based partitioning, a moisture deficit calculation, and an inert organic matter definition — that confound the structural comparison. SP1 and TP2 provide a cleaner complexity ladder in which each step adds precisely one structural element (a second pool and one transfer coefficient) while keeping the climate modifier, litter inputs, and likelihood architecture identical.

Core research question: Do divergent model projections of Finnish forest carbon sink saturation reflect genuine structural differences between models, or calibration artefacts? All models are calibrated against the same data with the same statistical framework; structural differences are designed to appear in residuals rather than being absorbed into error terms.

Pipeline split:

Script	Purpose
Data_work.R	Data preparation
run_{Model}_calibration.R	MCMC calibration, posterior samples
run_{Model}_predictive.R	Posterior predictive trajectories, obs vs pred
run_residual_analysis.R	Random forest residual analysis (model-agnostic)
run_multimodel_comparison.R	Cross-model metrics and overlay plots

2 Models

2.1 SP1 (Single-Pool)

SP1 is the simplest model in the suite. All litter inputs (NWL + FWL + CWL, all AWEN fractions collapsed to a single total flux J) enter one carbon pool C , which decomposes by first-order kinetics:

$$\frac{dC}{dt} = J(t) - \alpha \xi(t) C(t)$$

The steady-state pool size is $C^* = J/(\alpha\xi)$, and the annual step solution is exact:

$$C(t+1) = C_{ss}(t) + (C(t) - C_{ss}(t)) e^{-\alpha\xi(t)}$$

where $C_{ss}(t) = J(t)/(\alpha\xi(t))$ is the instantaneous steady state. Pure R implementation; no Fortran.

The climate modifier ξ uses the identical formula to Yasso07 (same functional form and data flow; see Section 2.3), with independently calibrated parameters β_1, β_2, γ .

2.2 TP2 (Two-Pool, ICBM-style)

TP2 follows the ICBM topology (Andren & Katterer 1997): all litter inputs enter the active pool A; the humus pool H receives only transferred mass from A. Using Yasso07 parameter notation:

$$\begin{aligned}\frac{dA}{dt} &= J(t) - \alpha_A \xi(t) A(t) \\ \frac{dH}{dt} &= p_H \alpha_A \xi(t) A(t) - \alpha_H \xi(t) H(t)\end{aligned}$$

where α_A , α_H are pool decomposition rates and p_H is the humification fraction (fraction of A decomposition entering H). This makes the TP2 parameter set a structural subset of Yasso07, facilitating direct comparison of pool-specific posteriors.

Analytical steady states:

$$A^* = \frac{J}{\alpha_A \xi}, \quad H^* = \frac{p_H J}{\alpha_H \xi}$$

Note that ξ cancels in H^* through the cascade, so $H^* = p_H \cdot A^* \cdot \alpha_A / \alpha_H$.

The transient step is solved exactly (not Euler-approximated) by integrating A's trajectory analytically as a time-varying input to H:

$$H(t+1) = H(t) e^{-k_H} + \frac{p_H J}{k_H} (1 - e^{-k_H}) + \frac{p_H k_A \Delta_A}{k_H - k_A} (e^{-k_A} - e^{-k_H})$$

where $k_A = \alpha_A \xi$, $k_H = \alpha_H \xi$, and $\Delta_A = A(t) - A_{ss}$. The degenerate case $k_A = k_H$ is handled via the L'Hôpital limit. Pure R implementation; no Fortran.

2.3 Yasso07

The Yasso07 model (Tuomi et al. 2009, 2011) describes SOC decomposition as first-order kinetics in five pools: A (acid-soluble), W (water-soluble), E (ethanol-soluble), N (non-soluble/lignin), and H (humus). Litter enters the A, W, E, N pools proportionally to AWEN fractions; material flows between AWEN pools (controlled by transfer fractions p_{ij}) and to H; H decomposes irreversibly.

The climate modifier is applied multiplicatively to all decomposition rates:

$$\xi(T, T_{\text{amp}}, P) = \frac{1}{4} \sum_{k=1}^4 \exp(\beta_1 T_k + \beta_2 T_k^2) \times (1 - \exp(\gamma P/1000))$$

where $T_1, \dots, T_4$ are quarterly temperatures derived from the annual mean T and seasonal amplitude T_{amp} via a Gaussian approximation of the sinusoidal cycle; P is annual precipitation (mm); and β_1 , β_2 , γ are free parameters. A woody decomposition size modifier scales decomposition rates for FWL/CWL:

$$h_S(d) = \min\{1, (1 + \delta_1 d + \delta_2 d^2)^{-|r|}\}$$

where d is woody litter diameter (cm). Model core implemented in Fortran; called via `.Fortran()`.

The SP1 and TP2 xi formula is identical to Yasso07 (same functional form, same data flow). Parameters β_1 , β_2 , γ are independently calibrated per model, using the same `compute_xi_yasso07()` R implementation from the Yasso07 wrapper. This ensures the climate response comparison is not confounded by different approximation schemes.

2.4 Yasso15 / Yasso20

Structurally identical AWEN pool structure as Yasso07, with pool-group-specific climate responses: separate $(\beta_1, \beta_2, \gamma)$ triplets for the AWE pool group, N pool, and H pool. Both Yasso15 and Yasso20 share the same Fortran core and the same 35-parameter vector structure. The sole computational difference is the temperature integration method: Yasso15 uses a four-point Gaussian approximation of the seasonal temperature cycle, while Yasso20 averages $\exp(\beta_1 T + \beta_2 T^2)$ over all 12 monthly temperatures directly.

Yasso15 includes leaching parameters $w_1, \dots, w_5$ (positions 17–21 of the parameter vector) representing dissolved organic carbon export. In Yasso20, these positions are retained for structural compatibility but are explicitly ignored by the Fortran routine (`mod5c20`, confirmed from source). In Yasso20, leaching is controlled by a fixed scalar argument rather than calibrated parameters. The original Yasso20 calibration (FMI GitHub repository, <https://github.com/YASSOmodel>) fixed w_1 – w_5 at zero; they are retained in `FIXED_RATE_NAMES` accordingly.

Calibration follows the same architecture as Yasso07 with an expanded parameter vector.

3 Parameter Specification — SP1

3.1 Free Parameters

SP1 has no fixed parameters: there are no published global calibrations to anchor the kinetic rate, so all parameters are learned from the Finnish data.

HIKET name	Physical meaning	Transform
<code>alpha</code>	Decomposition rate (yr^{-1})	log
<code>beta1</code>	Temperature response, linear	log
<code>beta2</code>	Temperature response, quadratic	unconstrained
<code>gamma</code>	Precipitation response	unconstrained
<code>sigma_init</code>	Initial-condition uncertainty	log
<code>sigma_input</code>	Global litter scaling	log

Total: 6 free parameters.

The posterior α from SP1 can be interpreted as the “effective mean residence time of the entire Finnish mineral forest SOC column under a single-pool assumption” — a meaningful summary statistic for comparison with the Yasso pool structure.

4 Parameter Specification — TP2

4.1 Free Parameters

Likewise, TP2 has no fixed parameters. The parameter set is a named subset of Yasso07, enabling direct comparison of pool-specific posteriors.

HIKET name	Physical meaning	Transform
alpha_A	A pool decomposition rate (yr^{-1})	log
alpha_H	H pool decomposition rate (yr^{-1})	log
p_H	Humification fraction ($A \rightarrow H$)	logit
beta1	Temperature response, linear	log
beta2	Temperature response, quadratic	unconstrained
gamma	Precipitation response	unconstrained
sigma_init	Initial-condition uncertainty	log
sigma_input	Global litter scaling	log

Total: 8 free parameters.

Note that `alpha_A` and `alpha_H` are **free** in TP2, unlike Yasso07 where the corresponding rates are fixed. This is necessary because there are no published global TP2 calibrations. Comparing TP2’s posterior `alpha_A` to Yasso07’s fixed α_A (and its posterior `alpha_H` to Yasso07’s α_H) is itself a structural finding: it reveals whether a two-pool model with Finnish data recovers pool timescales consistent with the Yasso07 global calibration.

5 Parameter Specification — Yasso07

5.1 Fixed Parameters

Six parameters are fixed at published values (Tuomi et al. 2009, 2011) and are not calibrated:

HIKET name	Published notation	Physical meaning
alpha_A, alpha_W, alpha_E, alpha_N	$\alpha_A, \alpha_W, \alpha_E, \alpha_N$	Pool decomposition rate constants (yr^{-1})
p_H	p_H	Fraction of decomposed mass entering humus
alpha_H	α_H	Humus decomposition rate (yr^{-1})

Rationale for fixing: Rate constants are well-constrained across the international Yasso literature. Calibrating them against Finnish SOC alone risks overfitting to Finnish climate and litter inputs. Structural differences between models in this intercomparison are encoded in flow fractions and climate response — the scientifically relevant comparison axes.

5.2 Free Parameters

20 parameters are jointly calibrated (18 physical + 2 auxiliary uncertainty):

Inter-pool flow fractions (12): $p_{AW}, p_{AE}, p_{AN}, p_{WA}, p_{WE}, p_{WN}, p_{EA}, p_{EW}, p_{EN}, p_{NA}, p_{NW}, p_{NE}$. Each donor pool i must satisfy $\sum_j p_{ij} \leq B = 1 - p_H$ (mass balance with humus formation).

Climate response (3): β_1 (temperature, linear; log-transformed), β_2 (temperature, quadratic; unconstrained), γ (precipitation; unconstrained).

Woody litter size modifier (3): δ_1 (linear diameter effect; unconstrained), δ_2 (quadratic diameter effect; **log-transformed**), r (power; **log-transformed**).

Auxiliary uncertainty parameters (2):

- σ_{init} : multiplicative initial-condition uncertainty. Added in quadrature to σ_{obs} at the first observation per plot. Captures error from assuming the 1985 SOC equals the analytical steady state (log-transformed; strictly positive).
- σ_{input} : global litter scaling multiplier. Applied to all litter inputs before model evaluation (steady state and transient run consistently). Captures allometric uncertainty in NFI litter estimates (\$ \$20–50% intrinsic uncertainty). Global (same multiplier for all plots, species, and years). Log-transformed; centred at 1 (no scaling).

6 Parameter Specification — Yasso15 / Yasso20

6.1 Fixed Parameters

Same six rate constants as Yasso07 ($\alpha_A, \alpha_W, \alpha_E, \alpha_N, p_H, \alpha_H$), fixed at the Yasso15/20 published values.

Yasso15 and Yasso20: leaching parameters w_1 – w_5 . Positions 17–21 of the parameter vector are explicitly marked “IGNORED IN THIS FUNCTION” in both the Yasso15 and Yasso20 Fortran sources. Leaching enters the model via an external scalar argument (**leac**); the HIKET R wrappers set this to zero, disabling leaching in both models. The w parameters are therefore inert in the HIKET calibration and are included in **FIXED_RATE_NAMES**. How the original FMI calibration used these parameters is unknown, as no R wrapper is provided in the Yasso15 repository; the non-zero w values in the published Yasso15 posterior may reflect a different calling convention or simply unidentified parameters.

Yasso20 additional fixed parameters: six transfer fractions are structurally fixed at zero in the original Yasso20 calibration (posterior SD ≈ 0 across all 10,000 repository samples). These are fixed in the HIKET calibration and not sampled:

$$p_{NW} = p_{AE} = p_{WE} = p_{NE} = p_{AN} = p_{EN} = 0$$

6.2 Free Parameters

Yasso15: 24 physical + 2 auxiliary uncertainty = 26 free parameters. **Yasso20:** 13 physical + 2 auxiliary uncertainty = 15 free parameters (structural zeros removed from free set).

Inter-pool flow fractions: same 12 as Yasso07 (Yasso15); subset for Yasso20 (only the 6 non-zero fractions: $p_{AW}, p_{WA}, p_{WN}, p_{EA}, p_{EW}, p_{NA}$).

Pool-group-specific climate response (9 for Yasso15; 9 for Yasso20):

HIKET name	Physical meaning
beta1, beta2, gamma	Temperature and precipitation response for AWE pools

HIKET name	Physical meaning
betaN1, betaN2, gammaN	Temperature and precipitation response for N pool
betaH1, betaH2, gammaH	Temperature and precipitation response for H pool

Woody litter size modifier (3): δ_1 , δ_2 , r as in Yasso07.

Auxiliary uncertainty parameters (2): σ_{init} , σ_{input} as in Yasso07.

7 Parameter Naming Conventions

The three Yasso model versions use different parameter naming conventions across publications, original calibration files, and this HIKET implementation. SP1 and TP2 use the HIKET names exclusively (no prior naming conflicts). The table below reconciles conventions for the non-fraction parameters across all Yasso sources.

HIKET code name	y07par_gui.csv	y15par.csv / rda	Fortran theta(i) (Y15/20)	Paper notation	Physical meaning
alpha_A	aA	aA	θ_1	α_A	A pool rate (yr ⁻¹)
alpha_W	aW	aW	θ_2	α_W	W pool rate (yr ⁻¹)
alpha_E	aE	aE	θ_3	α_E	E pool rate (yr ⁻¹)
alpha_N	aN	aN	θ_4	α_N	N pool rate (yr ⁻¹)
beta1	b1	b1	θ_{22}	β_1	Temp. linear (AWE)
beta2	b2	b2	θ_{23}	β_2	Temp. quadratic (AWE)
gamma	gamma	g	θ_{28}	γ	Precip. (AWE)
betaN1	—	bN1	θ_{24}	β_{N1}	Temp. linear (N)
betaN2	—	bN2	θ_{25}	β_{N2}	Temp. quadratic (N)
gammaN	—	gN	θ_{29}	γ_N	Precip. (N)
betaH1	—	bH1	θ_{26}	β_{H1}	Temp. linear (H)
betaH2	—	bH2	θ_{27}	β_{H2}	Temp. quadratic (H)
gammaH	—	gH	θ_{30}	γ_H	Precip. (H)
delta1	phi1	th1	θ_{33}	ϕ_1	Size modifier linear
delta2	phi2	th2	θ_{34}	ϕ_2	Size modifier quadratic

HIKET code name	<i>y07par_gui.csv</i>	<i>y15par.csv</i> / rda	Fortran theta(i) (Y15/20)	Paper notation	Physical meaning
r	r	r	θ_{35}	r	Size modifier power
alpha_H	aH	aH	θ_{32}	α_H	Humus rate (yr ⁻¹)
p_H	pH	pH	θ_{31}	p_H	Humus fraction
w1-w5	—	w1-w5	$\theta_{17}-\theta_{21}$	w_1-w_5	Leaching (Yasso15); ignored in Yasso20

Fortran index reference. The Fortran `theta(i)` column refers to the original Yasso15/20 Fortran subroutine (`mod5c20`) from the FMI GitHub repository (<https://github.com/YASSOmodel>), not the HIKET reimplementation. The HIKET wrapper reorders parameters internally via `ORIG_TO_LORENZO` before calling the Fortran.

Critical ordering note. The original Yasso15/20 Fortran and `.dat` files use the ordering $(\alpha, p_{ij}, w_{1:5}, \beta, \beta_N, \beta_H, \gamma, \gamma_N, \gamma_H, p_H, \alpha_H, \delta_1, \delta_2, r)$, placing leaching before climate parameters. The HIKET implementation reorders to $(\alpha, p_{ij}, \beta, \gamma, \beta_N, \gamma_N, \beta_H, \gamma_H, p_H, \alpha_H, \delta_1, \delta_2, r, w_{1:5})$. The bijective column remapping is documented in `Priors_model_matching.R` (vector `ORIG_TO_LORENZO`).

8 Parameter Transformations

The DEzs sampler operates in unconstrained $\mathbb{R}^n$ where n is 6 (SP1), 8 (TP2), 20 (Yasso07), 26 (Yasso15), or 15 (Yasso20). Physical parameters are recovered by applying `to_original()` to every proposal.

8.1 Stick-Breaking (flow fractions, Yasso models)

Each of the four donor columns (A, W, E, N) has three outflows satisfying:

$$p_{ij} \geq 0 \quad \forall j, \quad \sum_j p_{ij} \leq B$$

The stick-breaking transform converts a vector $v \in \mathbb{R}^3$ to a valid fraction vector p :

$$p_i = \sigma(v_i) \times r_i, \quad r_{i+1} = r_i - p_i, \quad r_1 = B$$

where $\sigma(v) = (1 + e^{-v})^{-1}$ is the logistic sigmoid. The inverse (physical $\rightarrow$ unconstrained) follows by sequential logit transformation of the normalised remainder fractions.

8.2 Logit Transform (p_H in TP2)

$$v = \log \frac{\theta}{1 - \theta}, \quad \theta = \sigma(v)$$

Applied to p_H in TP2 (single proportion in $(0, 1)$ with no budget constraint).

8.3 Log Transform

Applied to parameters that must be strictly positive:

$$v = \log(\theta), \quad \theta = e^v$$

Applied to: σ_{init} , σ_{input} , α (SP1), α_A , α_H (TP2), β_1 , β_{N1} , β_{H1} (temperature sensitivities, must be positive for boreal decomposition), δ_2 (size modifier quadratic, must be positive to prevent numerical failure), r (size modifier power; only $|r|$ is used in Fortran).

8.4 Unconstrained

No transformation: β_2 , β_{N2} , β_{H2} , γ , γ_N , γ_H , δ_1 , $w_{1:5}$.

8.5 The $\delta_2 > 0$ Requirement

The size modifier is $h_S(d) = \min\{1, (1 + \delta_1 d + \delta_2 d^2)^{-|r|}\}$. The expression inside the brackets must stay positive for all realistic woody litter diameters d (Finnish stumps: 2–30 cm). With $\delta_2 > 0$ the quadratic opens upward and stays positive for all $d \geq 0$ given the published values ($\delta_1 = -1.71$). With $\delta_2 < 0$ the quadratic opens downward, reaches zero at some diameter d_0 , and the expression becomes negative beyond that point. Raising a negative number to the non-integer power $-|r|$ is undefined in Fortran and propagates as NaN, which returns $-\infty$ to the likelihood.

This is not a HIKET-specific issue: the same formula appears verbatim in both the original Yasso15 and Yasso20 Fortran sources from the FMI repository. Any parameter set with $\delta_2 < 0$ would produce NaN in the original code when applied to woody litter larger than a few millimetres. The problem did not manifest in the original calibrations because those datasets apparently contained little variation in woody litter diameter, leaving δ_2 unidentified and allowing it to drift negative without penalty.

Applying a log transform to δ_2 in the HIKET calibration enforces $\delta_2 > 0$ throughout the MCMC. The published defaults ($\delta_2 = +0.86$ for Yasso07, $+1.27$ for Yasso15/20) confirm the physically correct sign.

The log transform on r is a minor complement: since only $-|r|$ enters the Fortran expression, the sign of r is irrelevant. Sampling in unconstrained space therefore wasted half the prior support on sign-equivalent values. Note that `YASSO07_DEFAULT_PARAMS` stores r as -0.307 (negative), consistent with the Fortran convention; the calibration script applies `abs()` to obtain the positive prior centre $+0.307$ required by the log transform. The Yasso15 and Yasso20 wrappers store $r = +0.257$ (positive), requiring no correction.

8.6 Jacobian Correction

When parameters are reparameterised (e.g. the sampler proposes $v = \log \delta_2$ instead of δ_2 itself), the probability density must be corrected for the stretching or compression introduced by the transformation. The log-Jacobian $\log |\partial \theta / \partial v|$ is added to the log-likelihood at every evaluation:

$$\log p(\theta \mid y) \propto \log p(y \mid \theta) + \log p(\theta) + \log \left| \frac{\partial \theta}{\partial v} \right|$$

For stick-breaking, the Jacobian factor per element is $\sigma(v_i)(1 - \sigma(v_i)) \times r_i$. For the log transform it is $e^v = \theta$. For the logit transform it is $\theta(1 - \theta)$. The Jacobian is added to the likelihood (not the prior) so that the BayesianTools prior object remains a standard Gaussian.

9 Prior Distribution

9.1 Philosophy

The prior is a multivariate normal in unconstrained space, independent across parameters:

$$p(v) = \prod_k \mathcal{N}(v_k; v_k^*, \sigma_k^2)$$

Three parameter groups receive qualitatively different prior treatment:

1. **Climate and size modifier parameters (Yasso models):** informative priors derived from the original global calibration datasets (Tuomi et al. 2008–2011) or from the published Yasso15/20 MCMC posteriors (FMI GitHub repository, <https://github.com/YASSOmode>). These parameters are well-constrained by external data; tight priors prevent degenerate compensation between parameters (particularly the γ – σ_{input} loop identified during initial runs) and prevent exploration of physically invalid regions.
2. **Transfer fractions (Yasso models):** weakly informative. Prior width $\sigma_k = 1.0$ in unconstrained stick-breaking space. The stick-breaking transform already guarantees that fractions are non-negative and sum to at most $B = 1 - p_H$ regardless of the prior width. Finnish data should freely inform the pool flow structure — the primary structural comparison axis between Yasso model versions.
3. **Kinetic parameters and climate response (SP1 and TP2):** weakly informative, with sigma on the log scale. No external calibration datasets are available for SP1 or TP2; the Finnish data alone informs these parameters. Prior centres for climate parameters (β_1 , β_2 , γ) are set to Yasso07’s published values as a physically plausible starting point. The kinetic rate α (SP1) is centred at $\alpha = 0.09 \text{ yr}^{-1}$, giving $C^* \approx 76 \text{ tC/ha}$ at Finnish conditions ($J \approx 2.4 \text{ tC/ha/yr}$, $\xi \approx 0.35$). For TP2, α_A and α_H are centred at the Yasso07 published values (0.73 and 0.0015 yr^{-1} respectively), and p_H at the Yasso07 value (0.028).
4. **Auxiliary uncertainty parameters (σ_{init} , σ_{input}):** moderately informative, with prior width 0.5 in log space for all models. σ_{init} is centred at 0.10 (10% initial-condition uncertainty); σ_{input} at 1.00 (no a priori litter scaling).

9.2 Prior Centres — Yasso07

Prior centres are the published parameter values from Tuomi et al. (2009, 2011), confirmed to agree with `YASSO07_DEFAULT_PARAMS` in the wrapper to within numerical precision. Decomposition rates in `y07par_gui.csv` are stored as positive absolute values; the wrapper correctly uses them with a negative sign in the transfer matrix. The parameter r is stored as -0.307 in `YASSO07_DEFAULT_PARAMS` (following the Fortran sign convention) but the log transform requires a positive prior centre; `abs(r)` is applied before constructing `best_x`.

9.3 Prior Centres — Yasso15

Prior centres for climate and size modifier parameters are the posterior means from the original Yasso15 global calibration (FMI repository, `Yasso15.dat`, 10,000 thinned MCMC samples after dropping the MAP row). These agree with the published MAP values in `y15par.csv` to four significant figures after applying the column remapping (`ORIG_TO_LORENZO`).

9.4 Prior Centres — Yasso20

Prior centres for climate and size modifier parameters are the MAP values from the official Ryassofortran R package (<https://github.com/YASSOmodel/Ryassofortran>, file `data/sample_parameters.rda`). These are used in preference to the posterior means from `Yasso20.dat` (10,000 samples) because the Ryassofortran package represents the FMI’s current validated parameter set for Yasso20. Cross-validation against `ParY20.dat` (repository MAP file) confirms that the two sources agree to numerical precision for all parameters, and that `sample_parameters.rda` and `Yasso20.dat` posterior means differ by less than 10% for all climate/size parameters (largest difference: $\beta_1 = 0.158$ vs. 0.167 from posterior mean).

9.5 Prior Widths — Yasso07

No MCMC posterior samples are available from the Yasso07 original calibration. Prior widths are set from two sources: (a) published 95% credible intervals from Tuomi et al. 2010 Table 4 for the size modifier parameters; (b) physical range arguments for the climate parameters.

Parameter	Prior width	Space	Basis
Flow fractions	1.0	unconstrained	Weakly informative
β_1	0.20	log	Physical range: $\exp(0.2 \times 10) \approx 7 \times$ modifier at $T = 10^\circ\text{C}$
β_2	0.05	physical	Prevents implausible modifier curvature
γ	0.30	physical	Keeps precip. modifier in $[0.3, 0.6]$ at Finnish P
δ_1	0.15	physical	Tuomi et al. 2010 Table 4: $\pm 0.16 \text{ cm}^{-1}$
δ_2	0.10	log	Tuomi et al. 2010 Table 4: $\pm 0.10 \text{ cm}^{-2}$
r	0.015	log	Tuomi et al. 2010 Table 4: ± 0.013

Parameter	Prior width	Space	Basis
σ_{init}	0.5	log	Physical 95% CI $\approx [0.04, 0.27]$
σ_{input}	0.5	log	Physical 95% CI $\approx [0.37, 2.72]$

9.6 Prior Widths — Yasso15 and Yasso20

For Yasso15/20, prior widths for climate and size modifier parameters are derived from the standard deviations of the published MCMC posterior samples from the FMI GitHub repository (`Yasso15.dat` and `Yasso20.dat`, 10,000 thinned samples each). For log-transformed parameters (β_1 , β_{N1} , β_{H1} , δ_2 , r) the width is $\text{SD}(\log \theta)$ across posterior samples; for unconstrained parameters it is $\text{SD}(\theta)$.

The γ_H prior width is capped at 1.5 in both models. At Finnish precipitation ($P \approx 600$ mm), $1 - \exp(\gamma_H \times 0.6) \approx 1$ for any $\gamma_H < -5$, meaning the H pool climate modifier is near its maximum regardless of γ_H . The large posterior SD for γ_H (3.99 for Yasso15, 1.39 for Yasso20) is consistent with this near-unidentifiability. The cap of 1.5 uses the better-constrained Yasso20 posterior as a reference.

Parameter	Yasso15 σ_k	Yasso20 σ_k	Space	Note
Flow fractions	1.0	1.0	unconstrained	Weakly informative
β_1	0.046	0.104	log	$\text{SD}(\log \beta_1)$ from posterior
β_2	0.00014	0.00054	physical	Very tight; well-identified
γ	0.070	0.146	physical	$\text{SD}(\gamma)$ from posterior
β_{N1}	0.105	0.062	log	$\text{SD}(\log \beta_{N1})$
β_{N2}	0.00008	0.00041	physical	
γ_N	0.155	0.035	physical	
β_{H1}	0.135	0.091	log	$\text{SD}(\log \beta_{H1})$
β_{H2}	0.00016	0.00009	physical	
γ_H	1.50	1.391	physical	Capped (near-unidentifiable at Finnish P)
δ_1	0.328	0.435	physical	$\text{SD}(\delta_1)$
δ_2	0.286	0.211	log	$\text{SD}(\log \delta_2)$
r	0.055	0.054	log	$\text{SD}(\log r)$
σ_{init}	0.5	0.5	log	Fixed
σ_{input}	0.5	0.5	log	Fixed

9.7 Prior Widths — SP1 and TP2

Parameter	Prior width	Space	Basis
α (SP1)	0.50	log	Wide; MRT 4–30 yr spanned
α_A (TP2)	0.50	log	Centred at Yasso07 value

Parameter	Prior width	Space	Basis
α_H (TP2)	0.50	log	Centred at Yasso07 value
p_H (TP2)	1.00	logit	Weakly informative
β_1	0.20	log	Same as Yasso07 for comparability
β_2	0.05	physical	Same as Yasso07
γ	0.30	physical	Same as Yasso07
σ_{init}	0.50	log	Same as Yasso
σ_{input}	0.50	log	Same as Yasso

9.8 Physical Sign Constraints

Several parameters must satisfy sign constraints from physical or mathematical requirements. These are enforced via log transforms rather than prior bounds, which would introduce hard edges incompatible with the Gaussian prior assumption.

Constraint	Parameters	Basis
> 0	$\beta_1, \beta_{N1}, \beta_{H1}$	Decomposition increases with temperature in boreal Finland.
> 0	δ_2	Size modifier $h_S(d) > 0$ for all Finnish diameters requires positive quadratic. Negative δ_2 causes NaN in Fortran for $d \gtrsim 0.4$ cm.
> 0	r	Only $- r $ enters Fortran. Sign is irrelevant; log transform avoids wasting prior support.
> 0	$\alpha, \alpha_A, \alpha_H$	Decomposition rates must be positive.
< 0	$\gamma, \gamma_N, \gamma_H$	$1 - \exp(\gamma P/1000) > 0$ requires $\gamma < 0$. Enforced by tight prior centred on negative published values.

9.9 Published Parameter Values at Prior Centres

Yasso07 (from `y07par_gui.csv`; Tuomi et al. 2009, 2011; r shown as $|r|$ — the prior centre):

Parameter	Value
β_1	0.09873
β_2	-\$0.00157
γ	-\$1.2717
δ_1	-\$1.7084
δ_2	+\$0.8586
r	+\$0.3068

Yasso15 (posterior means from `Yasso15.dat`; agree with `y15par.csv` to four significant figures):

Parameter	Value
β_1	0.0906
β_2	\$-\$0.000215
γ	\$-\$1.8090
β_{N1}	0.04878
β_{N2}	\$-\$0.0000792
γ_N	\$-\$1.1729
β_{H1}	0.03518
β_{H2}	\$-\$0.000208
γ_H	\$-\$12.541
δ_1	\$-\$0.4388
δ_2	\$+\$1.2684
r	\$+\$0.2569

Yasso20 (MAP from Ryassofortran `sample_parameters.rda`; <https://github.com/YASSOmodel/Ryassofortran>):

Parameter	Value
β_1	0.1580
β_2	\$-\$0.00200
γ	\$-\$1.4400
β_{N1}	0.17000
β_{N2}	\$-\$0.00500
γ_N	\$-\$2.0000
β_{H1}	0.06700
β_{H2}	\$+\$0.00000
γ_H	\$-\$6.9000
δ_1	\$-\$2.5500
δ_2	\$+\$1.2400
r	\$+\$0.2500

10 Likelihood

10.1 Observation Model

A multiplicative-normal (proportional-error) model is used:

$$\text{SOC}_{\text{obs},t} \sim \mathcal{N}(\hat{C}_t, s_t^2)$$

where the standard deviation scales with the predicted SOC:

$$s_t = \begin{cases} \hat{C}_t \sqrt{\sigma_{\text{obs}}^2 + \sigma_{\text{init}}^2} & \text{first observation per plot (1985)} \\ \hat{C}_t \sigma_{\text{obs}} & \text{subsequent observations (2006, 2024)} \end{cases}$$

Rationale for multiplicative-normal: A log-normal error model was used during a period when the data contained a $\times 10$ litter overestimate (since corrected in `Data_work.R`). With litter inputs in the correct boreal range (median ≈ 2.4 tC/ha/yr), a multiplicative-normal model is appropriate.

10.2 Fixed Observation Error

σ_{obs} is **fixed** at the observed coefficient of variation of SOC across all calibration plots:

$$\sigma_{\text{obs}} = \frac{\text{sd}(\text{SOC}_{\text{obs}})}{\text{mean}(\text{SOC}_{\text{obs}})} = 0.472$$

This value is **identical across all five models**. Fixing σ_{obs} ensures structural differences appear in residuals rather than being absorbed into the error term, which is critical for the intercomparison.

10.3 Log-Likelihood

For each plot i with n_i observations:

$$\ell_i(\theta) = \sum_{k=1}^{n_i} \log \mathcal{N}(\text{SOC}_{i,k}; \hat{C}_{i,t_{ik}}, s_{i,k}^2)$$

Total log-posterior in unconstrained space:

$$\mathcal{L}(v) = \sum_i \ell_i(\text{to_original}(v)) + \log \left| \frac{\partial \theta}{\partial v} \right|$$

If any plot returns $-\infty$, the entire proposal is rejected. The inner per-plot function is byte-compiled with `cmpfun()` for a ~ 10 – 20% speedup at the R level.

11 Per-Plot Likelihood Evaluation

Each likelihood evaluation executes three stages per plot in parallel via `mclapply`.

11.1 Stage 1: Environmental Modifier Time Series

For each year t in 1985–2023, `compute_xi()` computes the scalar modifier ξ_t from annual climate summaries (mean temperature, temperature amplitude, annual precipitation). The modifier is applied multiplicatively to all decomposition rates. For SP1 and TP2, the same `compute_xi_yasso07()` R function is used directly (no separate implementation), ensuring that any difference in fitted climate response reflects genuine structural differences between models, not formula discrepancies.

11.2 Stage 2: Steady-State Initialisation

The initial SOC pool distribution C^* is computed analytically from the closed-form steady state. For the Yasso models this is a matrix inversion $C^* = -A(\theta, \xi_{ss})^{-1} b(\theta)$; for SP1 and TP2 the steady states are explicit scalar/vector formulas.

The climate scalar used is $\xi_{ss} = \xi(\bar{T}, \bar{P})$, computed from mean climate over the **20-year steady-state window** (1985–2004).

Jensen’s inequality correction. Because ξ is nonlinear in T and P :

$$\xi(\bar{T}, \bar{P}) \neq \overline{\xi(T_t, P_t)}$$

Using the average of annual modifiers instead of the modifier at average climate introduces a 5–15% systematic bias in boreal conditions. The correct approach is enforced through the `compute_xi_mean()` interface: it averages T , T_{amp} , and P over the steady-state window and calls `compute_xi()` once.

Litter scaling. σ_{input} is applied to litter inputs in both the steady-state initialisation and the transient run, so the calibrated multiplier affects the initial condition and the trajectory consistently.

11.3 Stage 3: Transient Simulation

Starting from C^* , the model runs forward 1985–2023 using the annual ξ_t time series and actual annual litter inputs. Yasso models use a single Fortran call per plot; SP1 and TP2 use exact analytical solutions in R. Predicted SOC at observation years (1985, 2006, 2024) is extracted and compared to observations.

12 MCMC Sampling

12.1 Sampler

DEzs (Differential Evolution with snooker updates; ter Braak & Vrugt 2008), implemented in `BayesianTools` (Hartig et al. 2023). Selected for:

- No gradient requirements (black-box model functions).
- Robust performance on correlated, moderately high-dimensional posteriors.
- Population-based proposals that efficiently explore posterior ridges from compositional constraints.

12.2 Settings

Setting	Test (Mac)	Production (Puhti)
Chains	4	4
Iterations	10,000	50,000
Burn-in	2,000	10,000
DEzs jitter ε	0.001	0.001
Random seed	2025	2025

Burn-in fraction: 20%. Four chains give meaningfully tighter Gelman–Rubin bounds than three chains for negligible extra cost.

12.3 Parallelisation

Sequential-chains (current): chains run one after another (`lapply`), each using all available cores for the `mclapply` per-plot loop. On Puhti: 39 cores per chain (`parallelly::availableCores()`). On Mac: `detectCores() - 1`. SP1 and TP2 use the same architecture; because their per-plot evaluation is pure R (no Fortran call), individual evaluations are faster but the wallclock savings are modest given the R-level overhead.

Hybrid (future Puhti production): outer `lapply` replaced by `mclapply(mc.cores = 4)` with `CORES_PER_CHAIN = 90`, giving $4 \times 90 = 360$ of 384 cores. Nested forking is supported on Linux but not macOS, so sequential is mandatory for Mac development.

`parallel = FALSE` is passed to `createBayesianSetup()` so BayesianTools’ built-in parallelism is suppressed.

12.4 Starting Point

All chains start from `best_x = to_unconstrained(free_defaults)`. DEzs initialises its population from small random perturbations around `best_x` and rapidly disperses via differential-evolution proposals.

13 Pre-MCMC Sanity Check

Before launching MCMC, a forward simulation at the prior centre (published defaults) is run for 4 randomly selected plots:

1. **Numerical stability:** all predicted SOC finite and positive.
2. **Order-of-magnitude check:** predicted / observed $\in [0.1, 10]$ at all observation years.
3. **Likelihood check:** `ll_fn(best_x)` finite across all calibration plots.

Output: `{Model}_forward_at_defaults_{RUN_ID}.png` (2×2 panel). If the sanity check fails, MCMC is not launched.

14 Convergence Diagnostics

All diagnostics computed on post-burnin samples. $\hat{R}$ and ESS computed in unconstrained space; posterior summaries reported in physical space via `to_original()`.

14.1 Gelman–Rubin $\hat{R}$

$\hat{R}$	Interpretation
< 1.05	Good convergence
1.05–1.10	Marginal; extend chains

$\hat{R}$	Interpretation
> 1.10	Poor convergence; run not usable

Computed both per-parameter (univariate, `coda::gelman.diag()`) and jointly (multivariate $\hat{R}$).

14.2 Effective Sample Size

`coda::effectiveSize()`. Values < 200 indicate poor mixing.

14.3 Diagnostic Outputs

All files written to `Calibration_real_data/diagnostics/{Model}/:`

File	Content
<code>*_traces_*.png</code>	Trace plots per parameter, one line per chain
<code>*_marginals_fractions_*.png</code>	Prior vs posterior KDE for the flow fractions
<code>*_marginals_climate_aux_*.png</code>	Prior vs posterior KDE for climate and auxiliary parameters
<code>*_rhat_barplot_*.png</code>	$\hat{R}$ per parameter, coloured by convergence zone
<code>*_kl_divergence_*.png</code>	KL(posterior $\parallel$ prior) per parameter
<code>*_report_*.txt</code>	$\hat{R}$, ESS, posterior quantile table, chain health

Prior and posterior KDEs are evaluated on the same x-grid derived from the prior's support to avoid misleading compression.

15 Output Files

Written to `Calibration_real_data/runs/:`

File	Content	Used by
<code>{Model}_chains_{RUN_ID}.rds</code>	Raw BayesianTools chain objects (unconstrained)	Chain extension, re-diagnostics
<code>{Model}_posterior_{RUN_ID}.rds</code>	Post-burnin samples in physical space	Predictive script, downstream analysis
<code>{Model}_metadata_{RUN_ID}.rds</code>	Run config, R version, wallclock, summary stats	Reproducibility
<code>{Model}_inputs_{RUN_ID}.rds</code>	Per-plot structures: <code>climate_by_plot</code> , <code>inputs_by_plot</code> , <code>litter_means</code> , <code>obs_meta</code> , <code>plot_info</code>	Predictive script

The inputs RDS ensures exact provenance: the predictive script reads the same per-plot structures the calibration used, even if `Data_work.R` is subsequently updated.

16 Key Design Decisions

Architectural decisions locked in after discussion (April–May 2026):

1. **Multiplicative-normal likelihood.** Reverted from a log-normal workaround added when litter inputs had a $\times 10$ overestimate. With correct litter magnitudes, multiplicative-normal is the appropriate model.
2. **Fixed $\sigma_{\text{obs}} = 0.472$ across all models.** Ensures structural differences appear in residuals. A poorly fitting model cannot mask its inadequacy by calibrating a larger error variance.
3. **$\xi(\bar{T}, \bar{P})$, not $\xi(T_t, P_t)$, for steady state.** Jensen’s inequality correction; 5–15% bias in boreal conditions. Fixed at the `compute_xi_mean` interface level.
4. **STEADY_STATE_YEARS = 20.** Uses 1985–2004 climate for the steady-state scalar, pre-dating accelerated boreal warming.
5. **Fixed decomposition rates (Yasso models).** $\alpha_A, \alpha_W, \alpha_E, \alpha_N, p_H, \alpha_H$ fixed at published values. SP1 and TP2 have no published global calibrations, so all parameters are free.
6. **Log transform on δ_2 and r .** Enforces $\delta_2 > 0$ to prevent NaN in the Fortran size modifier and eliminates sign-redundancy in r .
7. **Informative priors for climate/size parameters (Yasso models).** Derived from published posteriors (Yasso15/20) or calibration confidence intervals (Yasso07). Yasso20 prior centres use the FMI Ryassofortran package MAP values (`sample_parameters.rda`) as the authoritative source, preferred over the `Yasso20.dat` posterior means.
8. **Yasso20 structural zeros fixed.** Eleven parameters with SD = 0 in the original Yasso20 calibration are fixed rather than sampled. Sampling near a hard boundary degrades DEzs mixing.
9. **SP1 and TP2 replace RothC.** RothC introduces conceptual differences (clay partitioning, moisture deficit, IOM) beyond pool count that confound the structural comparison. SP1 and TP2 provide a clean complexity spectrum (1 pool $\rightarrow$ 2 pools $\rightarrow$ 5 pools) in which each step adds one structural element while keeping all other modelling choices identical.
10. **SP1 and TP2 share the Yasso07 xi formula.** The same `compute_xi_yasso07()` R function is called directly in all three simple models. This ensures climate response comparisons are not confounded by approximation-scheme differences.
11. **ICBM topology for TP2.** All litter inputs enter the active pool A; H receives transfers only. This eliminates one free parameter (no input partitioning fraction) and is standard in Nordic forestry SOC models.
12. **Sequential chains.** Mandatory on macOS. Hybrid strategy (4×39 cores) available on Puhti.
13. **Global σ_{input} .** Single litter scaling multiplier. Per-species stratification is a possible extension.
14. **Three SOC observation years (1985, 2006, 2024).** The Komeetta campaign (2024) provides a third calibration target. The 2006–2024 stability constrains equilibrium decomposition rates independently of the 1985–2006 accumulation dynamics.

17 Software

Component	Details
R	≥ 4.3 ; Puhti: 4.5.2 (<code>r-env</code> Apptainer)
BayesianTools	Hartig et al. (2023); DEzs sampler
coda	Gelman–Rubin $\hat{R}$, ESS
compiler	<code>cmpfun()</code> byte-compilation
parallel	<code>mclapply()</code> per-plot parallelism
Fortran model cores	R CMD SHLIB; <code>.Fortran()</code> interface (Yasso07/15/20 only)
SP1 / TP2	Pure R; no Fortran compilation required
HPC	CSC Puhti; <code>small</code> partition; 1 node, 40 cores, 2 GB/core

Random seed: `set.seed(2025)` before chain initialisation (consistent across all model scripts). Mac
test plot subsampling: `set.seed(42)`.